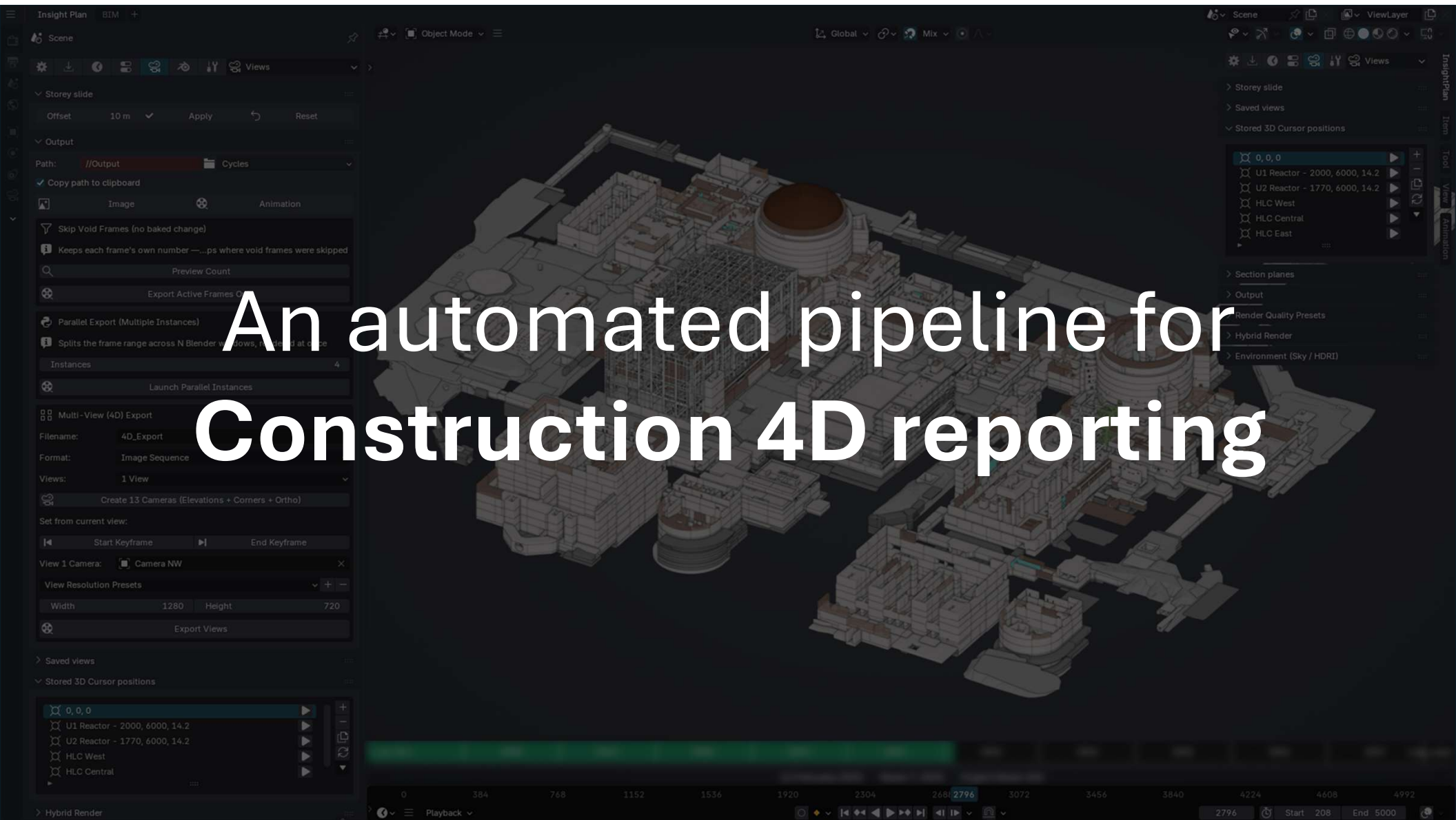


An automated pipeline for Construction 4D reporting



Speakers



Hing Pang Huang

Developer (Pythoneer)

at Hinkley Point C since November 2021



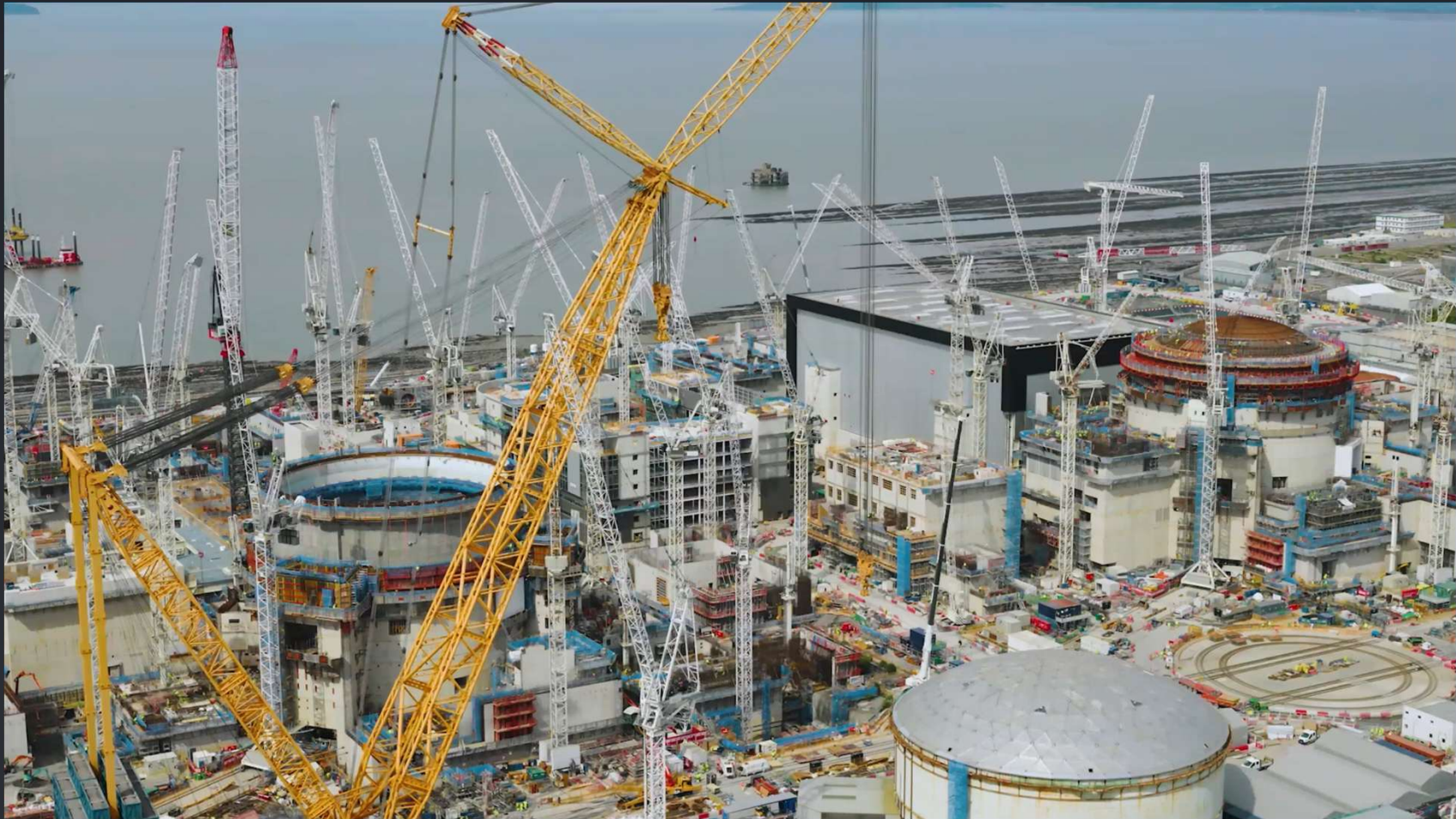
David Felix

Digital Engineer and Planner

at Hinkley Point C from 2017 to 2023
currently at Sizewell C

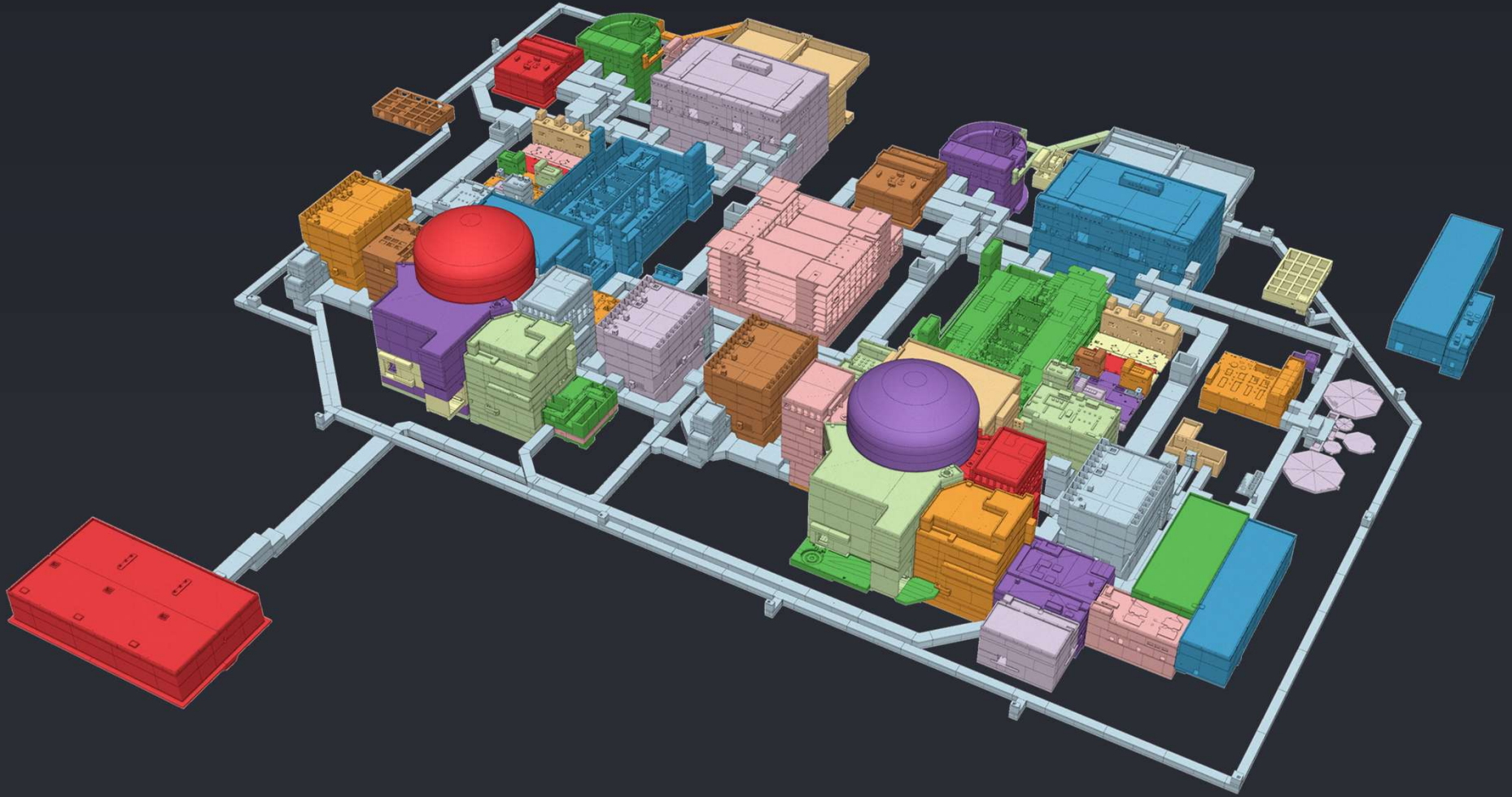


Nuclear Power Station - Hinkley Point C

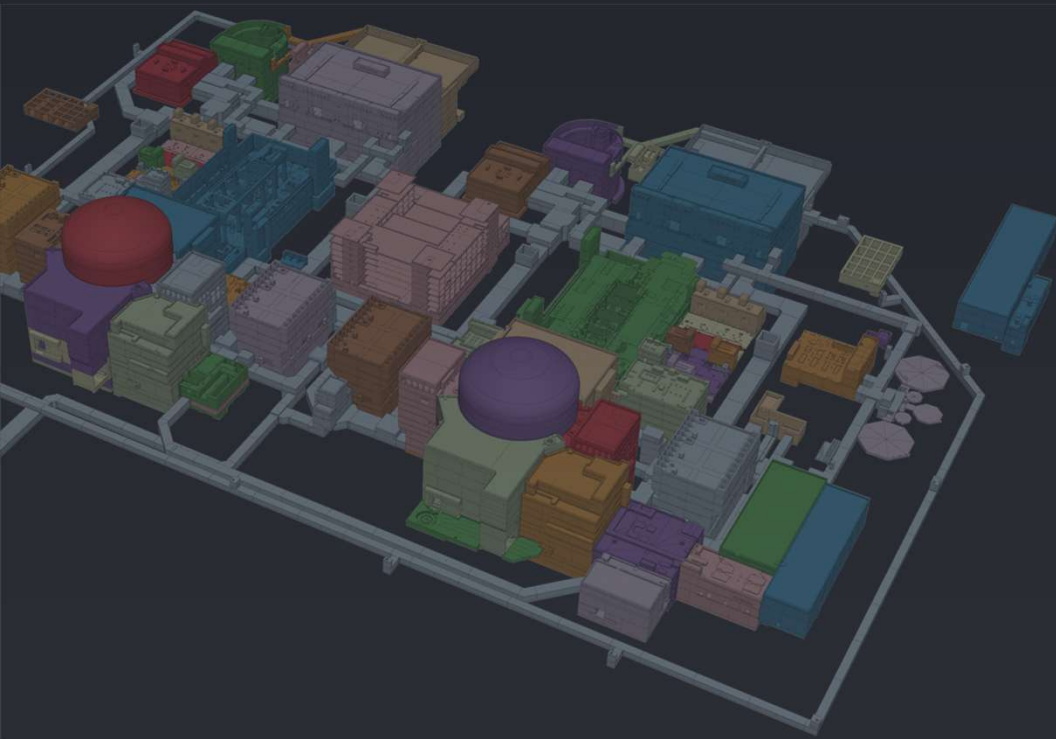


<https://www.youtube.com/watch?v=maWbEvM52Jo>

Nuclear Power Station - Hinkley Point C



Nuclear Power Station - Hinkley Point C



50 Buildings

55 Tower Cranes

25 Planners

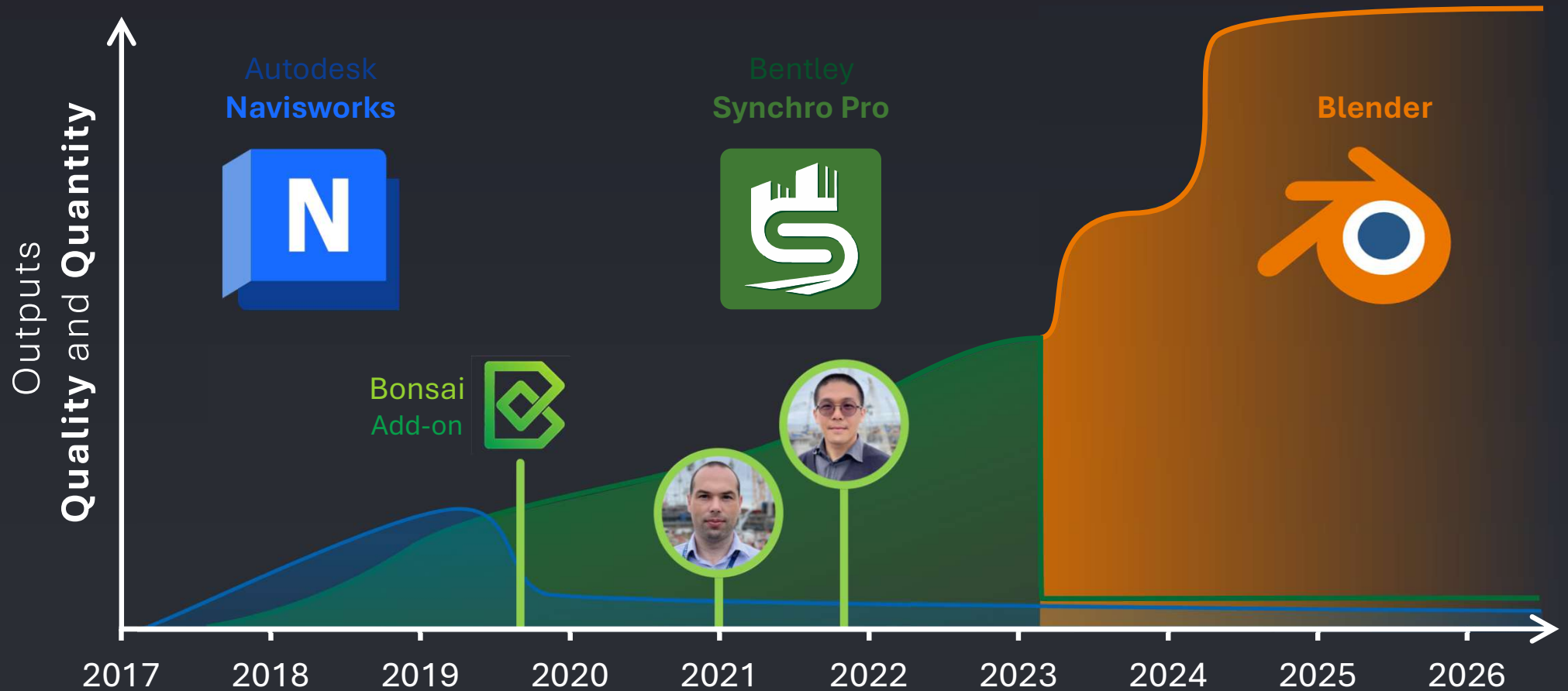
400 Models of Concrete

7200 Rooms

35000 Concrete Pours

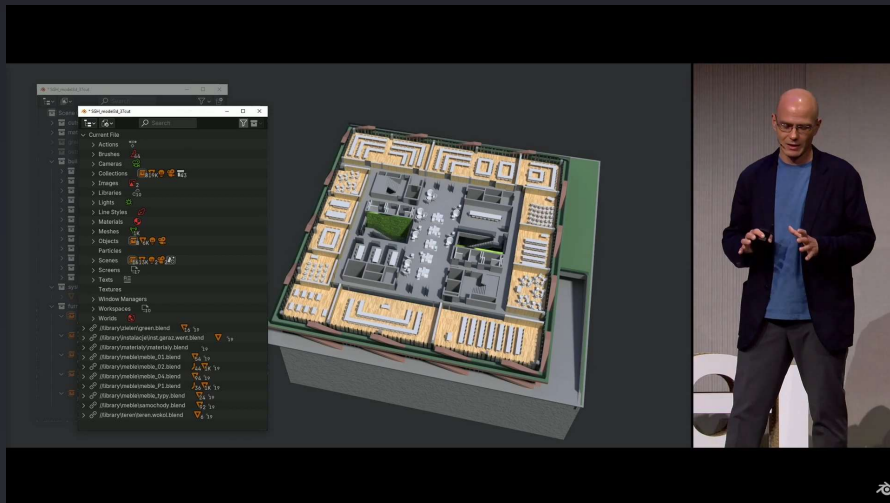
1 Digital Engineer - 4D

Software used for 4D during the years

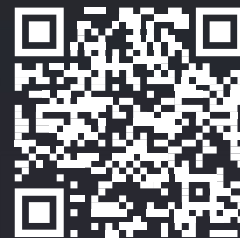


Bonsai references

(previously BlenderBIM Add-on)



Blending in architectural
design and on
construction site
by Paweł Czernecki



3D in Construction
by Daniel Bridges &
Renzo Caroca

3 key principles for successful 4D delivery

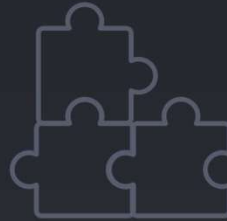
1

Start from the Output



2

Preparation is key

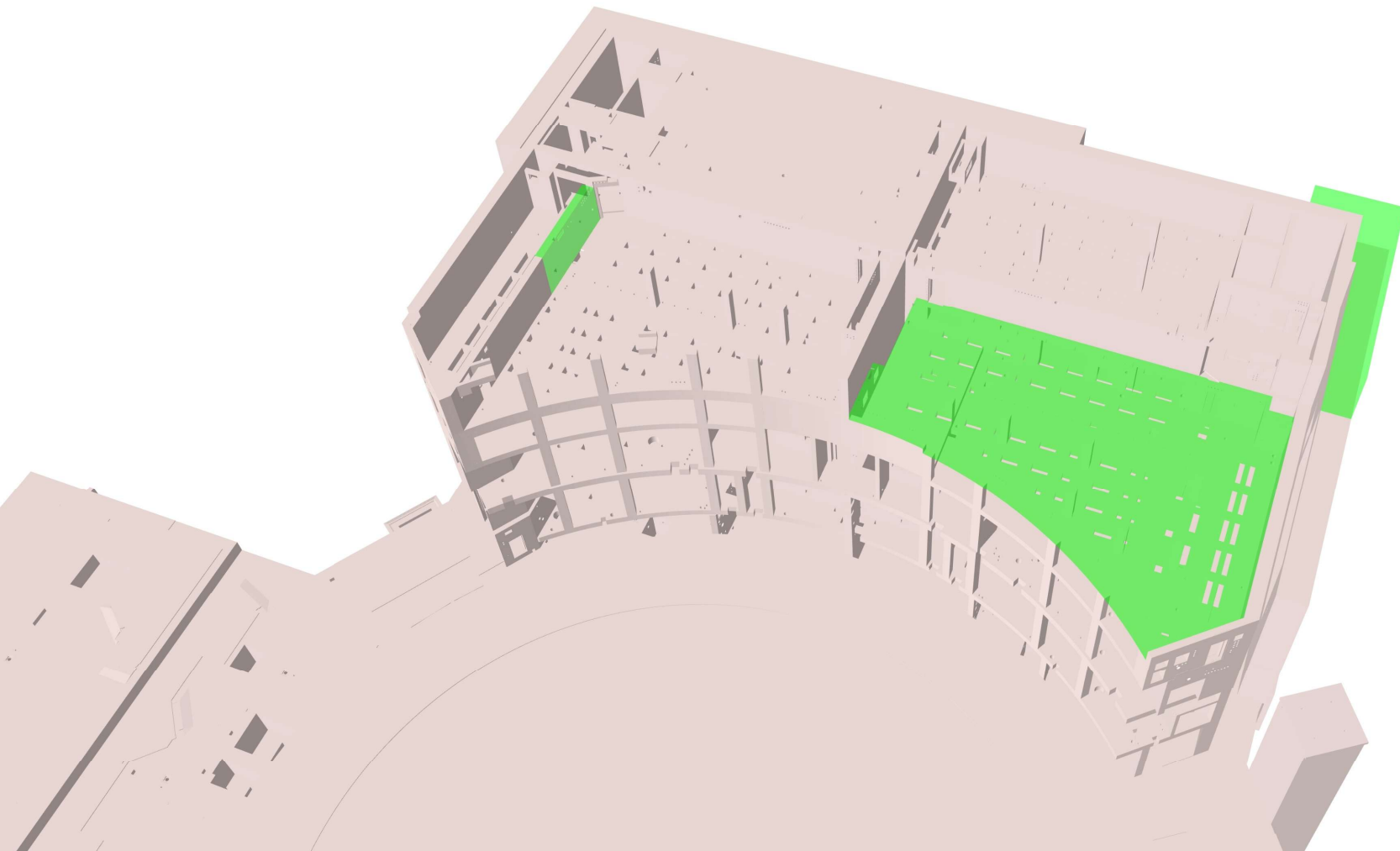


3

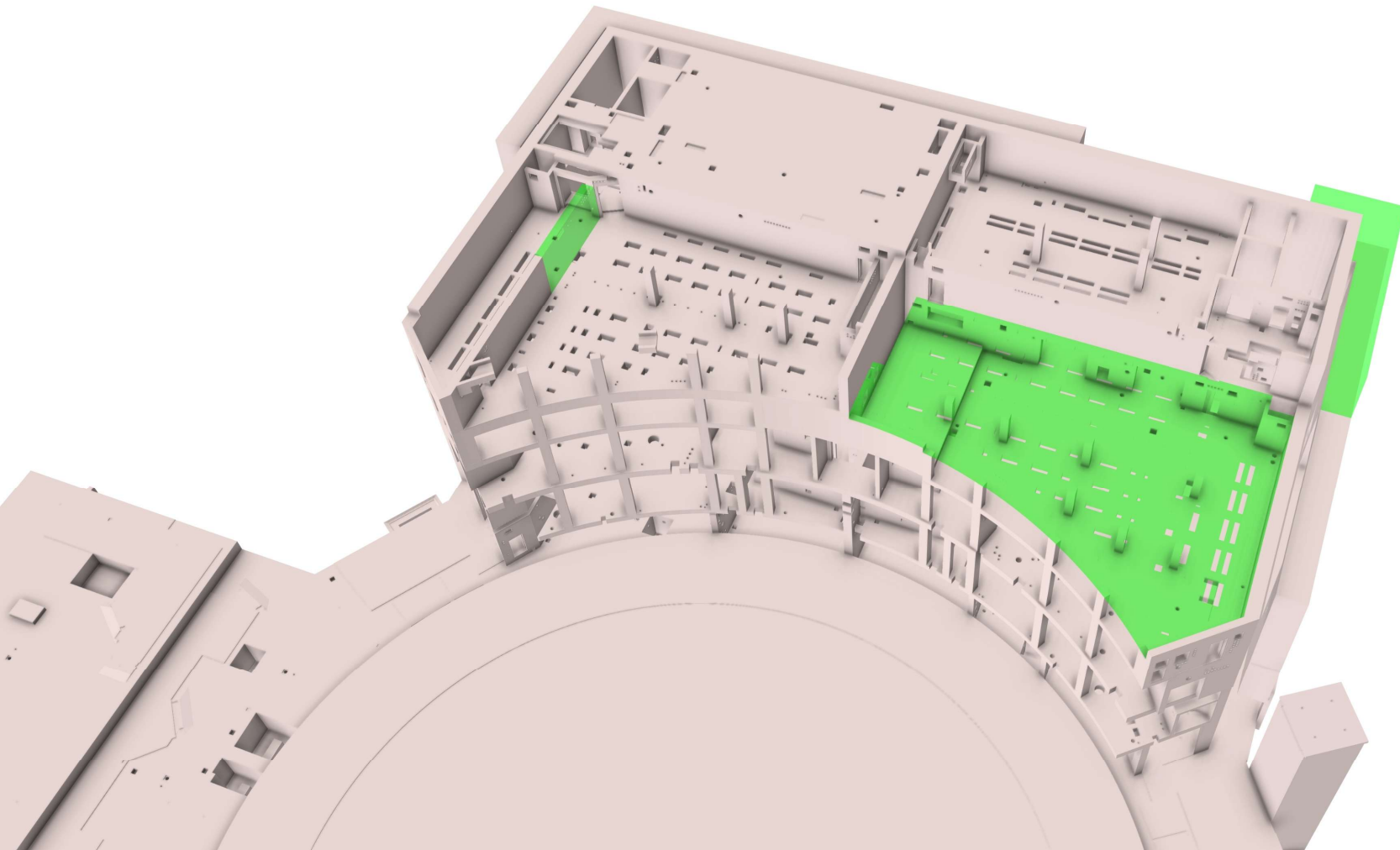
Ensure Scalability



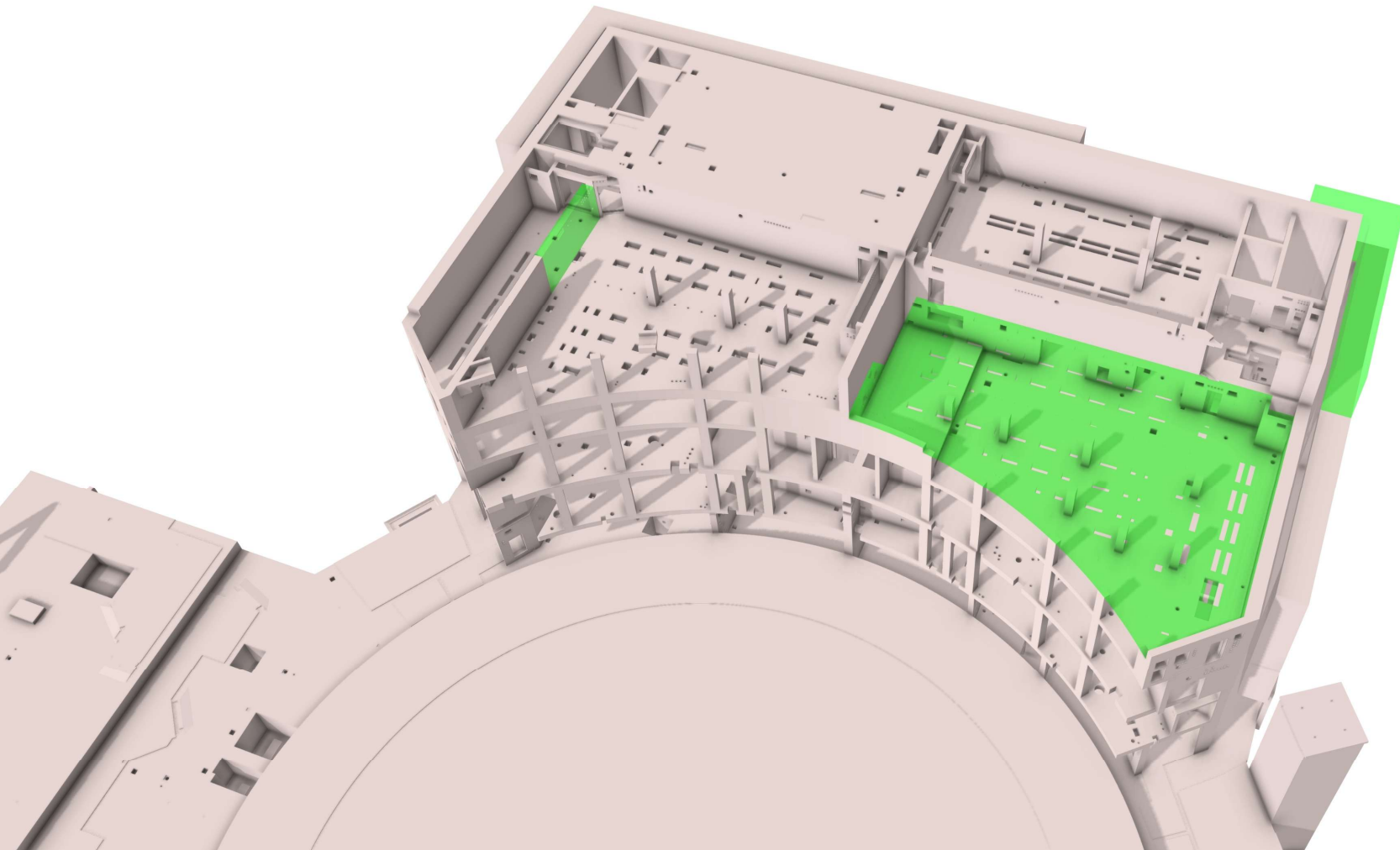
Average output **quality out of the box**



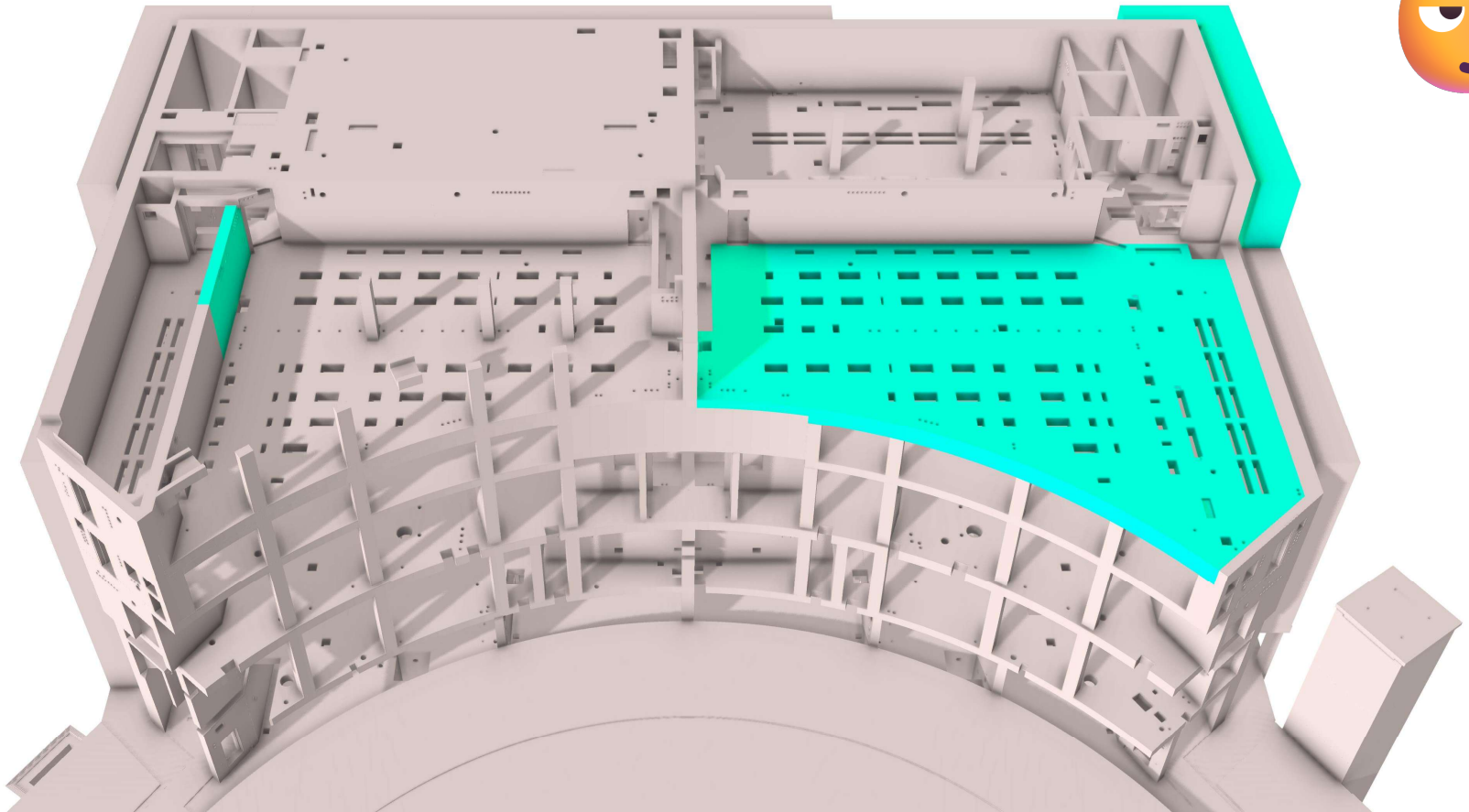
Let's turn on **Ambient Occlusion**



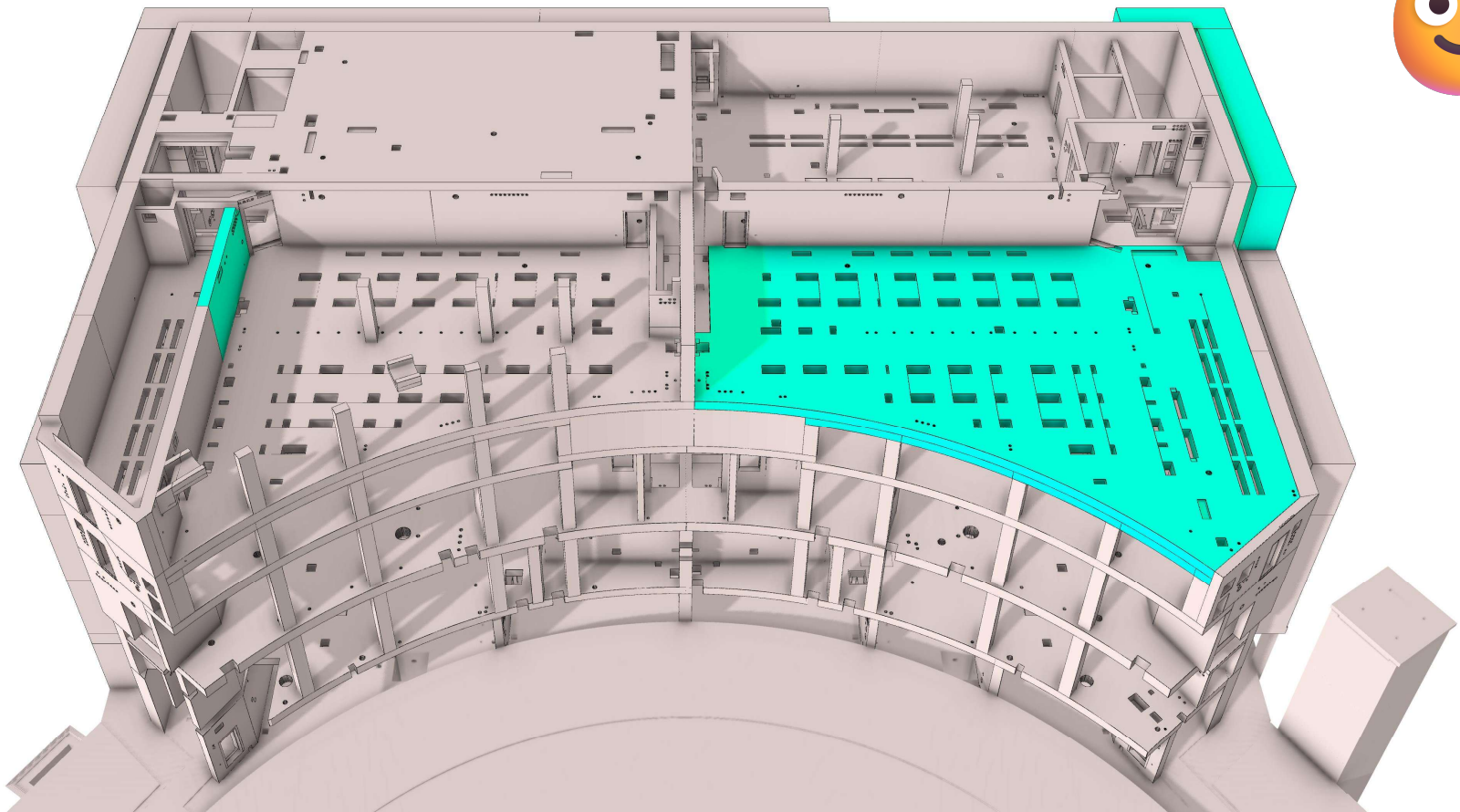
Let's turn on **Shadows**



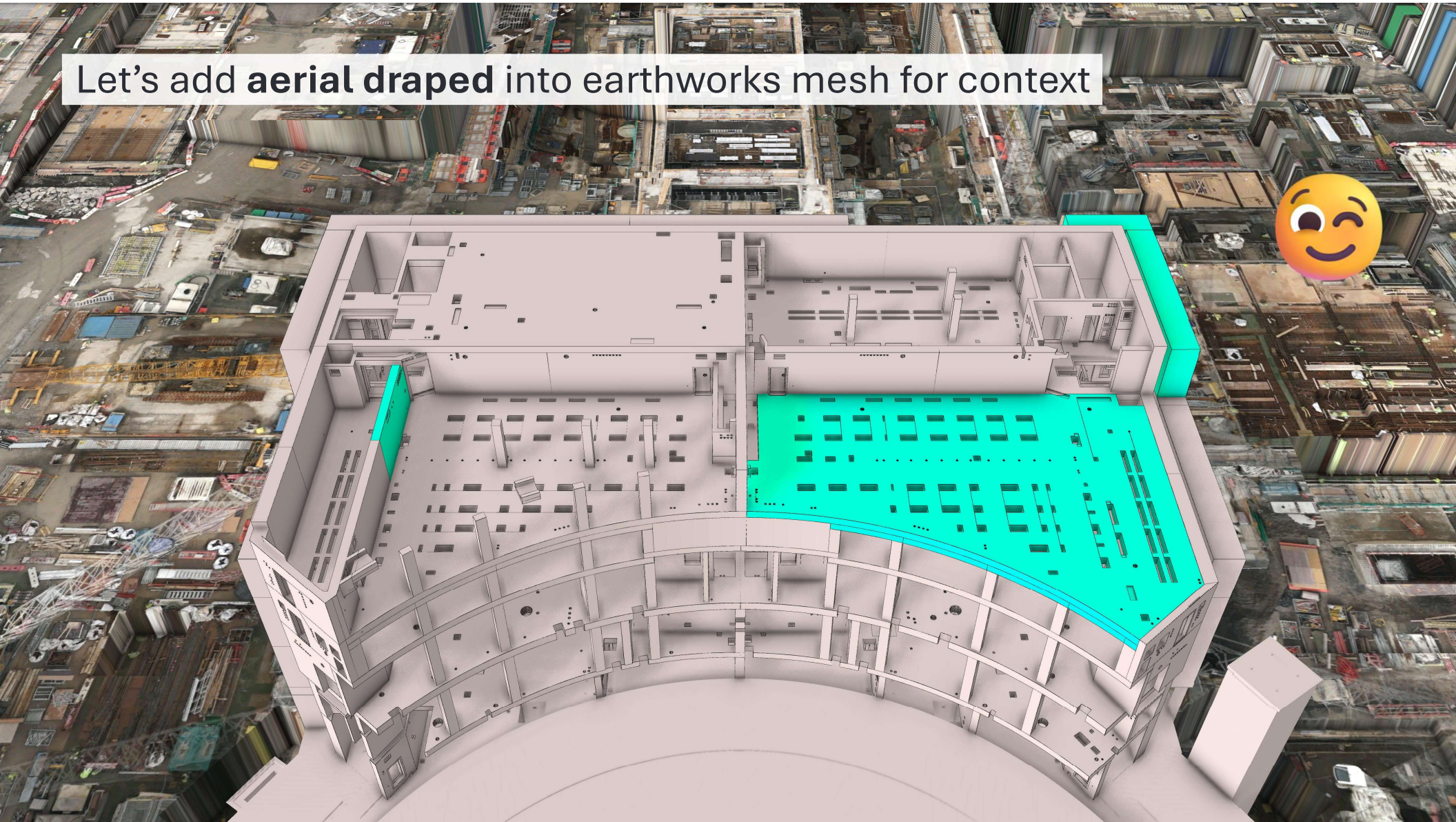
Standardise **Cameras** for consistency | Use custom appearance profiles



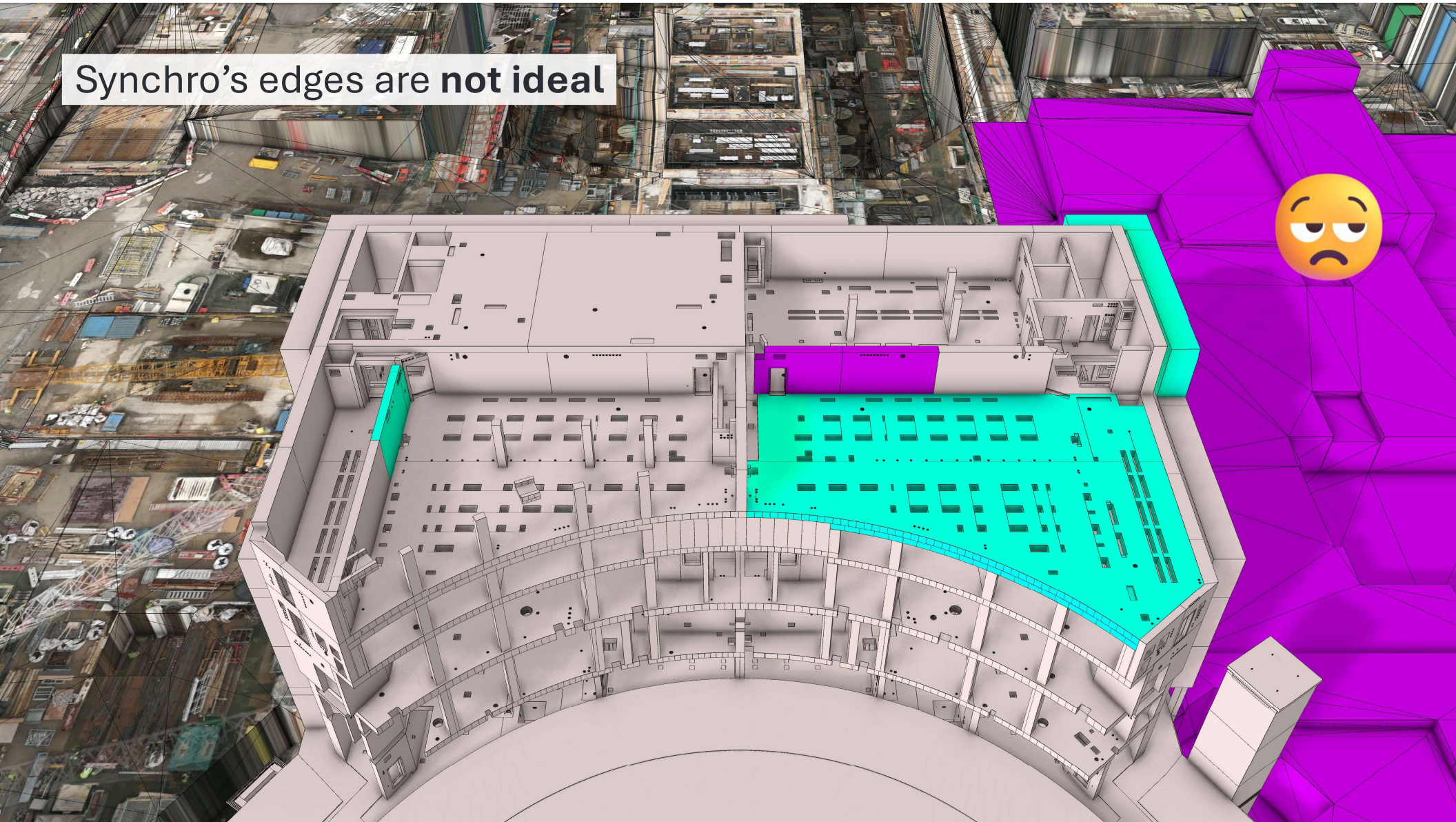
Let's add **contour lines** to see **construction joints**



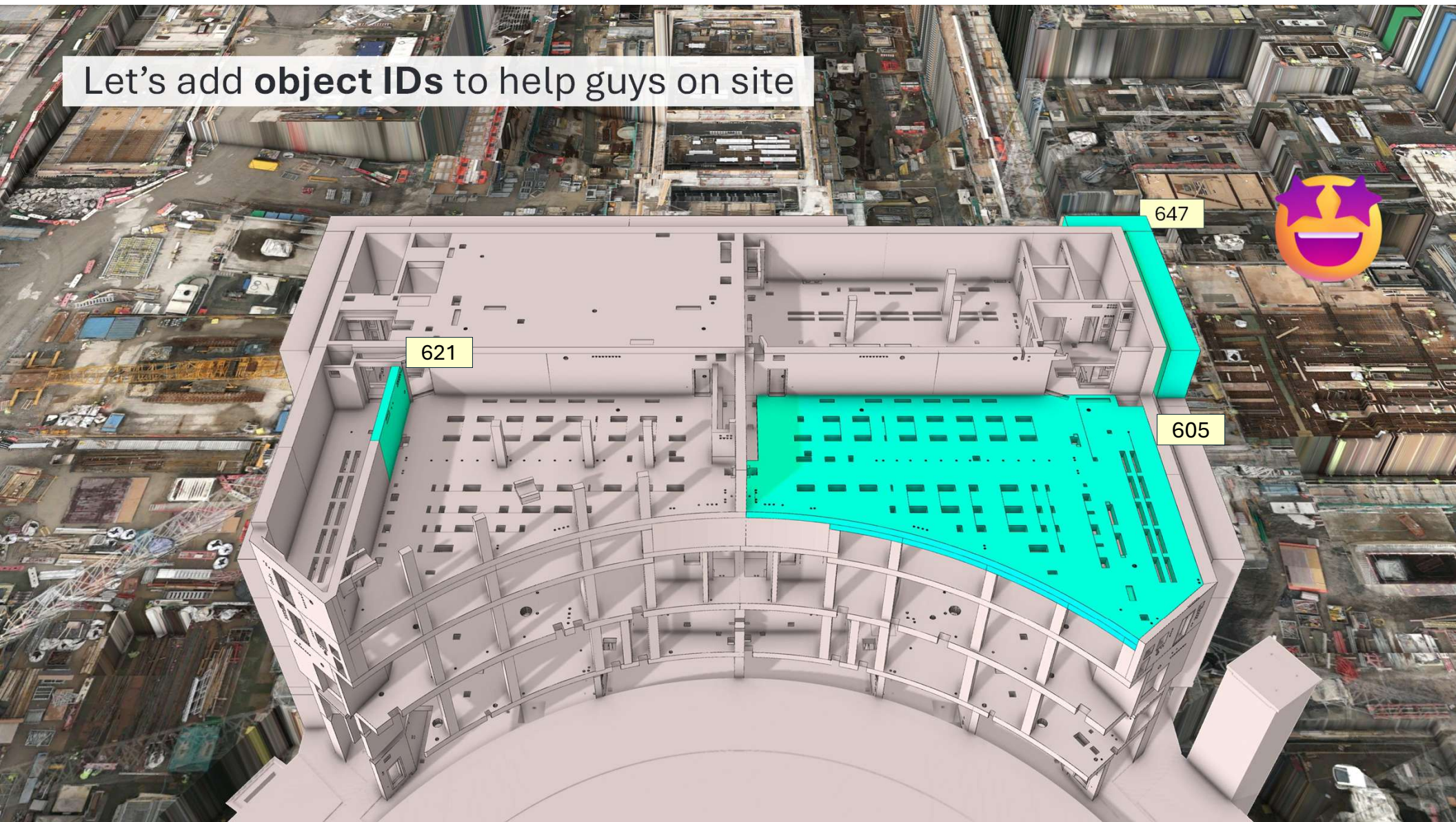
Let's add **aerial draped** into earthworks mesh for context



Synchro's edges are **not ideal**



Let's add **object IDs** to help guys on site



WEEK 7 NI-U1-HW

CONCRETE

LOOKING BACK

- Poured
- Not Poured
- Extra Poured

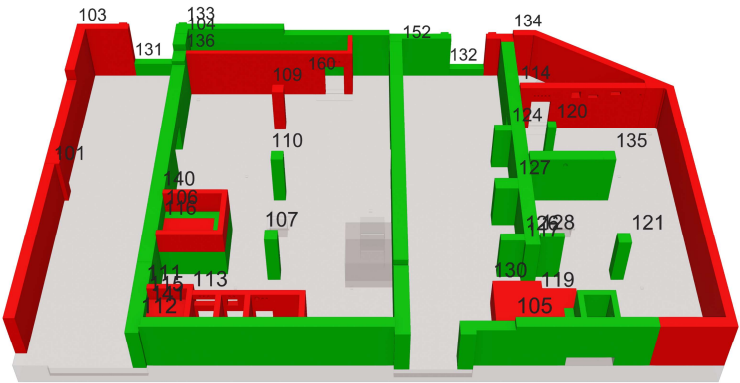


WEEK 7 NI-U1-HW

CONCRETE

LOOKING FORWARD

- On Target
- Not On Target
- Extra Planned

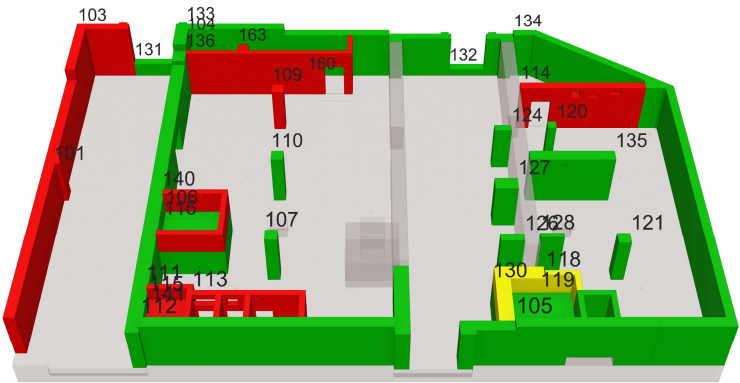


WEEK 7 NI-U1-HW

REINFORCEMENT

LOOKING FORWARD

- On Target
- Not On Target
- Extra Planned

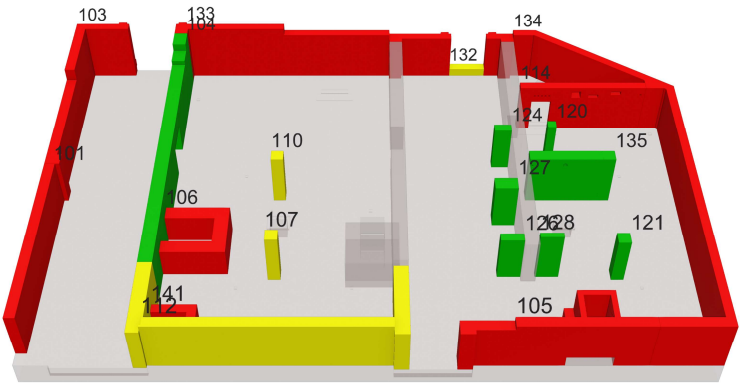


WEEK 7 NI-U1-HW

REINFORCEMENT

LOOKING BACK

- Installed
- Not Installed
- Extra Installed



4D Pipeline v1 – Hinkley Point C

1

Manually **export schedule** from Primavera P6 to an XER file

2

Schedule preparation using Python script to:

- 2.1. Convert XER tables into CSV files
- 2.2. Merge necessary data into CSV per project

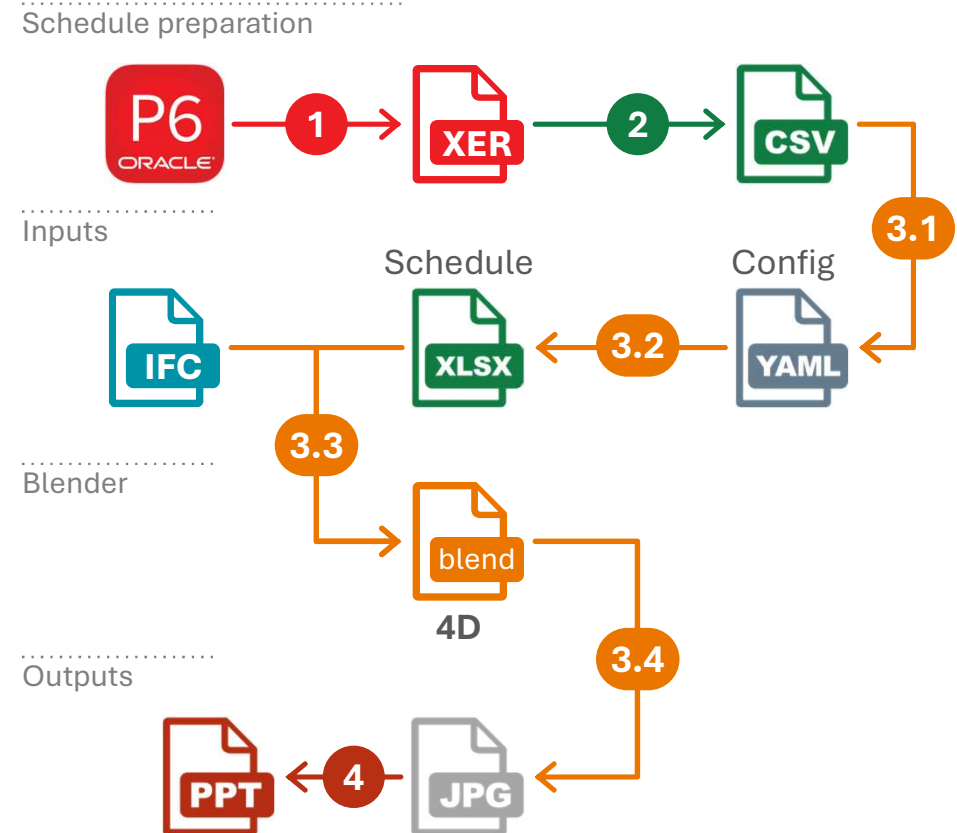
3

Animation logic and Blender using Python to:

- 3.1. Read configuration YAML file
- 3.2. Read CSV per project and prepare all 4D logic in Excel
- 3.3. Blender file preparation
 - a) Import IFC
 - b) 4D Keyframing based on external Excel
- 3.4. Export images

4

PowerPoint with macro imports all images



1 => N

How we scale

- Simplification & Progress reporting.
- Replicate the complete environment to users.
- Centralized Channel for updates and feedbacks.
 - Git Repo for update distribution.
 - Group Chat for feedback and announcement

Traceability

- Record Input Files and parameters. i.e. .blend file
- Logging for runtime parameter collection
- Able to replicate the runtime environment from collected data
- Remote debugging for detailed troubleshooting.

4D Pipeline v2 – Sizewell C pilot

1

Manually **export schedule** from Primavera P6 to an **XER** or **XML** file, schedules in **Excel** also work

2

Blender file preparation using Blender addon that uses Python to:

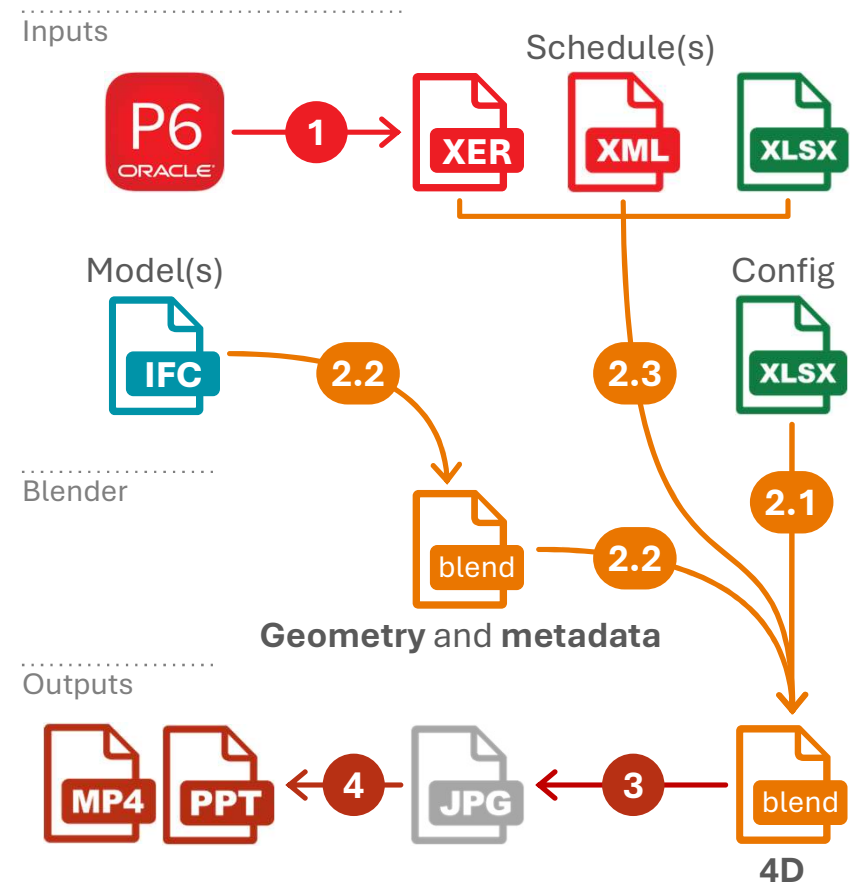
- 2.1. Read Configuration XLSX file
- 2.2. Import IFCs - one .blend per model, then append into a master .blend
- 2.3. Read schedules XER, XML or XLSX, parses and stores directly inside the .blend
 - a) Automatch tasks with objects
 - b) 4D keyframing based on embedded data

3

Export images

4

PowerPoint with macro imports all images, or turn the images into video with a Python script



4D Pipelines comparison

	v1	v2
1 Execution	Headless batch only	Headless batch , or interactively as a Blender add-on
2 Schedule inputs	XER only	XER, XML and XLSX
3 Schedule storage	Outside Blender, in XLSX	Inside Blender
4 Model inputs	One IFC per run	Multiple IFCs , merged into one combined scene
5 Object metadata	Not cached, re-read from source each run	Cached as <i>[PropertySet]Name</i> custom properties
6 Feature scope	4D sequencing for reporting	Same as v1 plus extra functionalities for Planning : colouring, storey explode, Multi-programme sync, Automatch, Gantt Chart
7 Animation speed	Fixed per build: a new cadence needs regenerating and rebuilding	Change anytime in Blender: pick day/week/month/year, then re-animate

Insight Plan

BIM

+

Scene

⚙️

📄

🔄

🔗

🎥

🔧

🕒

4D

Schedule Import

F:\K10 and Tower Cranes.xml

?

Not synced

F:\F06_28JUN26.xer

?

Not synced

F:\B1 Backfill.xlsx

?

Not synced

F:\Granular Backfill.xlsx

?

Not synced

Not synced

Sync All

Automatch All

Create Animation

Frames

1

1

Day

Start Date

2017-01-01

Finish Date

2028-05-02

Guess

Start Frame

1

Finish Frame

5000

Calculate Frame Range

Speed Type

Frame-based

Create Animation

Jump to Date

Current: 2023-05-04 (frame 2796)

Jump

Timeline Bar

Timeline HUD

Add from Clipboard/Text

Enter P6 XML/XER file paths (one per line):

"F:\K10 and Tower Cranes.xml"

"F:\F06_28JUN26.xer"

"F:\B1 Backfill.xlsx"

"F:\Granular Backfill.xlsx"

"F:\K10 and Tower Cranes.xml""F:\F06_28JUN26.xer""F:\B1 Backfill.xlsx""F:\Granular Backfill.xlsx"

Tip: Copy paths from Windows Explorer or a text file

OK

Cancel

Day

Real Duration Unit

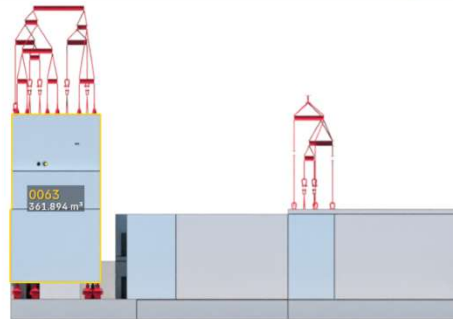
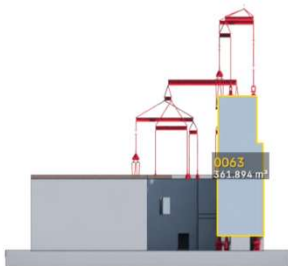
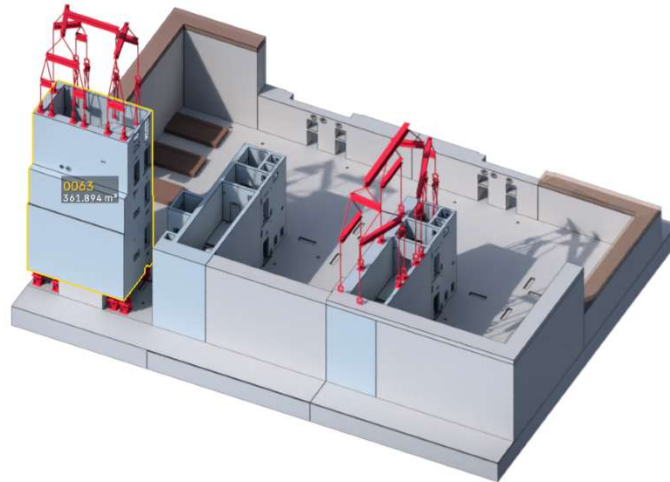
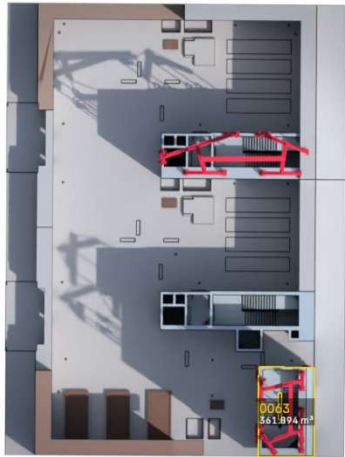
Day

Week

Month

Year

4D Pipeline v2 – New Visual Capabilities



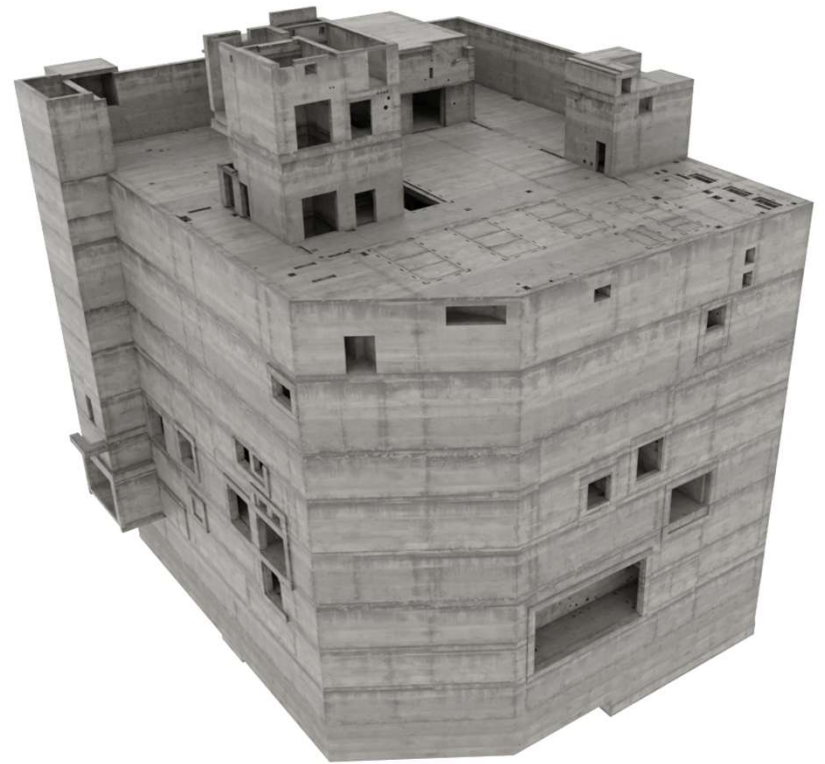
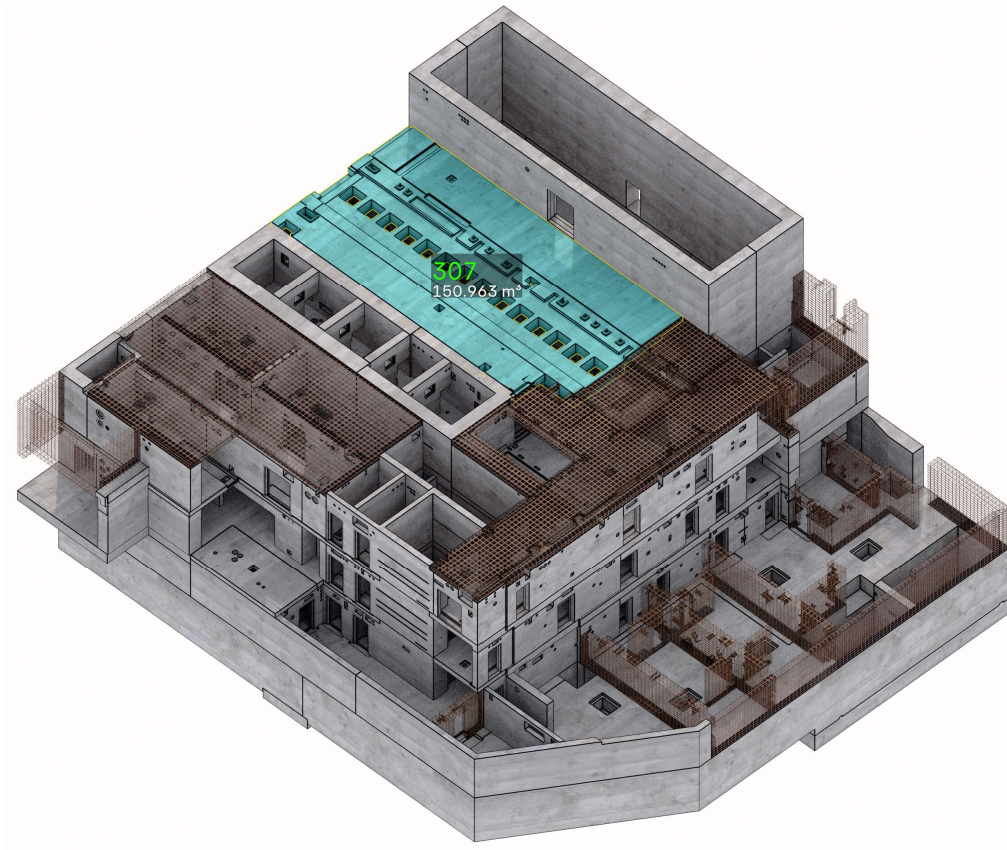
Presentation

- Animated camera from close-up to full building
- Multi-view render: 1, 2, 3 or 4 views
- Custom labels on active tasks
- Growth effect (e.g. dropping precast, left to right on a slab)
- Animated Monthly aerial

Model content

- Realistic materials
- Temporary works and secondary steel
- Neighbouring buildings, backfill and site logistics

4D Pipeline v2 – Realistic materials



Key Takeaways

- ✓ **Blender sits in the right place for what we need now**
Scheduled 4D image sequences, on modest hardware, easy to deploy and a Python/headless workflow we already know how to automate and customise.
- ✓ **Automation demands discipline**
Models and P6 schedules must match precisely and remain accurate.
- ✓ **Python + IfcOpenShell allow manipulation of IFC models**
Fix common model issues programmatically with Python before 4D sequencing.
- ✓ **Multiprocessing cuts run times**
Process multiple IFC/Blender files, or renderings in parallel.
- ✓ **Faster workflows = time for more creative & precise outputs**
Cutting times dramatically opens space to experiment with alternatives, improve accuracy, and create richer, more decision-useful 4D models.

